Six math clues · answer key

Electrons

Age 8 / Grade 2 math. Regroup ten ones as one ten for addition, or trade one ten for ten ones for subtraction. Welcome a sketch or spoken explanation.

Math 1 · 45

7 + 8 = 15 ones: 1 ten and 5 ones. Then 2 + 1 + 1 = 4 tens. 27 + 18 = 45 beads.

Math 2 · 36

64 is 5 tens and 14 ones. 14 − 8 = 6 ones; 5 − 2 = 3 tens. 64 − 28 = 36 beads.

Math 3 · 55

6 + 9 = 15 ones: 1 ten and 5 ones. Then 3 + 1 + 1 = 5 tens. 36 + 19 = 55 dots.

Math 4 · 35

82 is 7 tens and 12 ones. 12 − 7 = 5 ones; 7 − 4 = 3 tens. 82 − 47 = 35 bits.

Math 5 · 83

6 + 7 = 13 ones: 1 ten and 3 ones. Then 4 + 3 + 1 = 8 tens. 46 + 37 = 83 clips.

Math 6 · 39

95 is 8 tens and 15 ones. 15 − 6 = 9 ones; 8 − 5 = 3 tens. 95 − 56 = 39 stickers.

Age 8 / Grade 2 math. Six number-crunch tasks, all within 100, with two-digit operands and actual regrouping. Quantities describe craft supplies and model marks, not actual electron counts or measurements from art.

Reading & writing · answer key

Electrons

Reading · 1

Some electrons transfer from the wool to the balloon when the materials rub and separate.

Reading · 2

Electrons can move from one fabric to another. One can gain extra negative charge, while the other loses electrons and becomes positive. Opposite charges pull together.

Reading · 3

Both involve electrons carrying negative charge. A closed circuit supports current through a loop; static cling can follow electrons moving between materials and leaving a charge imbalance.

Writing

Accept a clear command spoken, dictated, drawn, or written. Example: “Rub the balloon on your sweater, then hold it near the paper bits.”

Age 8, Grade 3 reading, Grade 2 math. A grown-up leads every demo. Keep fingers out of outlets, and keep hands dry around plugs and switches. Use a balloon, sweater, paper bits, and a tray, or simply notice laundry cling. Keep balloon pieces and model beads with the grown-up. Use a battery-and-lamp activity only with a grown-up and a ready-made snap circuit or kit; leave wall outlets closed. The atom is a simplified spatial model, not a scale view or circular planetary orbit. Electrons have far less mass than protons and neutrons; the common hydrogen nucleus has no neutron. Charge marks in the diorama stand for an imbalance across many atoms, not one atom or measured particle counts. Its pieces exchange electrons without creating any or moving the positive nuclei. Static attraction can also involve polarization: a neutral object’s charges shift slightly without changing its total charge. Real transfer and cling depend on materials and humidity. The wire scene shows a low-voltage complete circuit; a lamp transfers energy, not a supply of used-up electrons.

Welcome speech, dictation, invented spelling, or writing. One command is enough. Example: “Rub the balloon on your sweater, then hold it near the paper bits.”

Four charge clues

Electrons

Quiz 1. What charge does an electron carry?

An electron carries negative electric charge.

Quiz 2. Which parts are packed in the nucleus of this atom model?

The model’s nucleus holds protons and neutrons. Electrons move around it.

Quiz 3. What can rubbing a balloon on wool do?

Rubbing can transfer some electrons from wool to the balloon. The balloon gains negative charge, and the wool loses electrons.

Quiz 4. What do opposite electric charges do?

Opposite charges pull toward each other. Like charges push apart.

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